

Complex Python

Lesson #5

By: Alan Alappatt

Table Of Contents

Classes and Instances 01

Self and Init 02

Inheirtance 03

Classes and Instances

A class is a blueprint. An instance is the actual item in the computer's memory. Python will make a new memory block for that object.

```
class Student:  2+ usages
    pass

student_a = Student()
student_b = Student()
```

Self and Init

Self is used to refer to a specific object created in the code. It's a temporary tag for the current object.
__init__ is a built-in function called an initializer.
username temporarilty holds the name we put in.
Next line permanently attaches the name from username into that specific object. Now it permanently remembers its own name.

```
class Account:  1 usage
    def __init__(self, username):
        self.username = username


user = Account("Alex")
print(user.username)
```

Activity Time!

Inheirtance

the parentheses tell us that Admin is a subclass of User. Admin will inheirit the attributes of the User class. The next line overrides the parent's init method(adds permissions). Super() is another built-in function that finds the parent class. It brings the stuff from the User class(email) to the Admin class.

```
class User:  1 usage
    def __init__(self, email):
        self.email = email

class Admin(User):
    def __init__(self, email, permissions):
        super().__init__(email)
        self.permissions = permissions
```

Activity Time!

Thank You!

Any Questions?